

Assignment 4 Solutions

Modern Cryptography (2)

Problem 1. Let

$$u = (3, 7), \quad v = (5, 10),$$

and let

$$L = \{au + bv : a, b \in \mathbb{Z}\}.$$

First, the norms of the basis vectors are

$$\|u\| = \sqrt{3^2 + 7^2} = \sqrt{58}, \quad \|v\| = \sqrt{5^2 + 10^2} = \sqrt{125} = 5\sqrt{5}.$$

Now we determine the shortest nonzero vectors in L . Any lattice vector has the form

$$(x, y) = a(3, 7) + b(5, 10) = (3a + 5b, 7a + 10b).$$

From this, we compute

$$y - 2x = (7a + 10b) - 2(3a + 5b) = a,$$

and

$$7x - 3y = 7(3a + 5b) - 3(7a + 10b) = 5b.$$

Hence, if $(x, y) \in L$, then $7x - 3y \equiv 0 \pmod{5}$, i.e.

$$2x - 3y \equiv 0 \pmod{5}.$$

Since $-3 \equiv 2 \pmod{5}$, this is equivalent to

$$2x + 2y \equiv 0 \pmod{5},$$

and because 2 is invertible modulo 5, we get

$$x + y \equiv 0 \pmod{5}.$$

Conversely, if $(x, y) \in \mathbb{Z}^2$ satisfies $x + y \equiv 0 \pmod{5}$, then

$$a = y - 2x \in \mathbb{Z}, \quad b = \frac{7x - 3y}{5} \in \mathbb{Z},$$

so $(x, y) = au + bv \in L$. Therefore,

$$L = \{(x, y) \in \mathbb{Z}^2 : x + y \equiv 0 \pmod{5}\}.$$

We now look for the smallest Euclidean norm among nonzero integer vectors satisfying $x + y \equiv 0 \pmod{5}$. A nonzero vector of norm 1 would have to be $(\pm 1, 0)$ or $(0, \pm 1)$, but none of these satisfies $x + y \equiv 0 \pmod{5}$. The vectors

$$(1, -1), \quad (-1, 1)$$

do satisfy the condition, and each has norm

$$\sqrt{1^2 + (-1)^2} = \sqrt{2}.$$

Hence the shortest vectors in L are

$$\boxed{\pm(1, -1)}.$$

Problem 2. Let

$$v_1 = (1, 1, 1), \quad v_2 = (1, 2, 3), \quad v_3 = (1, 0, 1).$$

We perform Gram–Schmidt orthogonalization.

Set

$$v_1^* = v_1 = (1, 1, 1).$$

Next,

$$\mu_{21} = \frac{\langle v_2, v_1^* \rangle}{\langle v_1^*, v_1^* \rangle} = \frac{1 \cdot 1 + 2 \cdot 1 + 3 \cdot 1}{1^2 + 1^2 + 1^2} = \frac{6}{3} = 2.$$

Thus

$$v_2^* = v_2 - \mu_{21}v_1^* = (1, 2, 3) - 2(1, 1, 1) = (-1, 0, 1).$$

Now compute

$$\mu_{31} = \frac{\langle v_3, v_1^* \rangle}{\langle v_1^*, v_1^* \rangle} = \frac{1 \cdot 1 + 0 \cdot 1 + 1 \cdot 1}{3} = \frac{2}{3}.$$

Also,

$$\mu_{32} = \frac{\langle v_3, v_2^* \rangle}{\langle v_2^*, v_2^* \rangle} = \frac{(1, 0, 1) \cdot (-1, 0, 1)}{(-1)^2 + 0^2 + 1^2} = \frac{0}{2} = 0.$$

Therefore,

$$v_3^* = v_3 - \mu_{31}v_1^* - \mu_{32}v_2^* = (1, 0, 1) - \frac{2}{3}(1, 1, 1) = \left(\frac{1}{3}, -\frac{2}{3}, \frac{1}{3}\right).$$

So the Gram–Schmidt orthogonalized vectors are

$$\boxed{v_1^* = (1, 1, 1), \quad v_2^* = (-1, 0, 1), \quad v_3^* = \left(\frac{1}{3}, -\frac{2}{3}, \frac{1}{3}\right)}.$$

One may also scale the last vector and write an orthogonal basis as

$$(1, 1, 1), \quad (-1, 0, 1), \quad (1, -2, 1).$$

Problem 3. We prove that the multi-secret version of decision-LWE is hard.

Let

$$\mathcal{H}_0 = (a, a^T s_1 + e_1, a^T s_2 + e_2, \dots, a^T s_t + e_t),$$

where

$$a \xleftarrow{\$} \mathbb{Z}_q^n, \quad s_1, \dots, s_t \xleftarrow{\$} \mathbb{Z}_q^n, \quad e_1, \dots, e_t \leftarrow \chi$$

are all sampled independently.

Let

$$\mathcal{H}_t = (a, u_1, \dots, u_t),$$

where

$$a \xleftarrow{\$} \mathbb{Z}_q^n, \quad u_1, \dots, u_t \xleftarrow{\$} \mathbb{Z}_q$$

are all sampled independently. This is the uniform distribution over

$$\mathbb{Z}_q^n \times \mathbb{Z}_q^t.$$

For each $i \in \{0, 1, \dots, t\}$, define the hybrid distribution

$$\mathcal{H}_i = (a, u_1, \dots, u_i, a^T s_{i+1} + e_{i+1}, \dots, a^T s_t + e_t),$$

where all variables are independent and uniformly/randomly sampled as above. Thus $\mathcal{H}_0$ is exactly the multi-secret LWE distribution, and $\mathcal{H}_t$ is exactly uniform.

We now claim that every two adjacent hybrids are computationally indistinguishable.

Claim. For every $i \in \{1, \dots, t\}$, the distributions $\mathcal{H}_{i-1}$ and $\mathcal{H}_i$ are computationally indistinguishable, assuming decision-LWE is hard.

Proof of the claim. Fix any $i \in \{1, \dots, t\}$. Suppose there exists a PPT distinguisher $\mathcal{D}$ such that

$$|\Pr[\mathcal{D}(\mathcal{H}_{i-1}) = 1] - \Pr[\mathcal{D}(\mathcal{H}_i) = 1]|$$

is non-negligible. We construct a PPT distinguisher $\mathcal{B}$ for the standard decision-LWE problem.

The input to $\mathcal{B}$ is a sample (a, b) , where either

$$b = a^T s + e \quad \text{for } s \xleftarrow{\$} \mathbb{Z}_q^n, e \leftarrow \chi,$$

or

$$b \xleftarrow{\$} \mathbb{Z}_q.$$

Using this challenge, $\mathcal{B}$ generates a tuple

$$(a, z_1, \dots, z_t)$$

as follows:

- for each $j < i$, sample $z_j = u_j \xleftarrow{\$} \mathbb{Z}_q$;
- set $z_i = b$;
- for each $j > i$, sample $s_j \xleftarrow{\$} \mathbb{Z}_q^n$ and $e_j \leftarrow \chi$, and let

$$z_j = a^T s_j + e_j.$$

Then $\mathcal{B}$ outputs whatever $\mathcal{D}$ outputs on input

$$(a, z_1, \dots, z_t).$$

If $b = a^T s + e$ is an LWE sample, then

$$(a, z_1, \dots, z_t)$$

is distributed exactly as $\mathcal{H}_{i-1}$.

If b is uniform in $\mathbb{Z}_q$, then

$$(a, z_1, \dots, z_t)$$

is distributed exactly as $\mathcal{H}_i$.

Therefore, any non-negligible advantage of $\mathcal{D}$ in distinguishing $\mathcal{H}_{i-1}$ from $\mathcal{H}_i$ gives a non-negligible advantage for $\mathcal{B}$ in distinguishing decision-LWE from uniform, contradicting the hardness of decision-LWE.

This proves the claim. □

Now, since t is polynomially bounded and each adjacent pair $\mathcal{H}_{i-1}, \mathcal{H}_i$ is computationally indistinguishable, the hybrid lemma implies that

$$\mathcal{H}_0 \approx_c \mathcal{H}_t.$$

That is, the distribution

$$(a, a^T s_1 + e_1, \dots, a^T s_t + e_t)$$

is computationally indistinguishable from the uniform distribution.

Hence the multi-secret form of decision-LWE is hard.

Problem 4. We prove that the dual Regev public-key encryption scheme is IND-CPA secure, assuming the decision-LWE problem is hard.

Let

$$\Delta := \left\lfloor \frac{q}{2} \right\rfloor.$$

We first note the correctness of decryption. For a ciphertext

$$c_1 = A^T r + e_1, \quad c_2 = r^T h + e_2 + \mu \Delta,$$

the decryptor computes

$$d = c_2 - c_1^T s.$$

Since $h = As$, we obtain

$$d = r^T As + e_2 - (A^T r + e_1)^T s + \mu \Delta = e_2 - e_1^T s + \mu \Delta \pmod{q}.$$

Thus, if the total error

$$\eta := e_2 - e_1^T s$$

has magnitude much smaller than $q/4$, then d is much closer to 0 when $\mu = 0$, and much closer to Δ when $\mu = 1$, so decryption is correct.

We now prove IND-CPA security by a sequence of games.

Game 0. This is the real IND-CPA experiment. The challenger samples

$$A \xleftarrow{\$} \mathbb{Z}_q^{n \times m}, \quad s \xleftarrow{\$} \{0, 1\}^m, \quad h = As,$$

and gives the public key

$$pk = (A, h)$$

to the adversary. On challenge messages $\mu_0, \mu_1 \in \{0, 1\}$, the challenger picks $\beta \xleftarrow{\$} \{0, 1\}$ and returns a real encryption of μ_β .

Game 1. This game is identical to Game 0, except that the public-key component h is sampled uniformly:

$$A \xleftarrow{\$} \mathbb{Z}_q^{n \times m}, \quad h \xleftarrow{\$} \mathbb{Z}_q^n.$$

The challenge ciphertext is still generated as

$$c_1 = A^T r + e_1, \quad c_2 = r^T h + e_2 + \mu_\beta \Delta.$$

Claim 1: Game 0 and Game 1 are statistically indistinguishable.

The only difference between the two games is the distribution of (A, h) : in Game 0 we have (A, As) with $s \xleftarrow{\$} \{0, 1\}^m$, while in Game 1 we have (A, u) with $u \xleftarrow{\$} \mathbb{Z}_q^n$.

Consider the hash family

$$\mathcal{F} = \{f_A : \{0, 1\}^m \rightarrow \mathbb{Z}_q^n\}_{A \in \mathbb{Z}_q^{n \times m}}, \quad f_A(s) = As.$$

This is a universal hash family, and s has min-entropy m . Therefore, by the Leftover Hash Lemma, if

$$m \geq n \log_2 q + \omega(\log \lambda),$$

then the distribution

$$(A, As)$$

is statistically close to

$$(A, u), \quad u \xleftarrow{\$} \mathbb{Z}_q^n.$$

Hence the adversary's distinguishing advantage changes by at most a negligible amount:

$$|\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_1 = 1]| \leq \text{negl}(\lambda).$$

Game 2. This game is identical to Game 1 except that the challenge ciphertext is chosen uniformly at random:

$$A \xleftarrow{\$} \mathbb{Z}_q^{n \times m}, \quad h \xleftarrow{\$} \mathbb{Z}_q^n,$$

and on challenge messages μ_0, μ_1 , after choosing $\beta \xleftarrow{\$} \{0, 1\}$, the challenger samples

$$u_1 \xleftarrow{\$} \mathbb{Z}_q^m, \quad u_2 \xleftarrow{\$} \mathbb{Z}_q$$

and returns

$$(c_1, c_2) = (u_1, u_2 + \mu_\beta \Delta).$$

Claim 2: Game 1 and Game 2 are computationally indistinguishable under decision-LWE.

Suppose there exists a PPT adversary $\mathcal{A}$ that distinguishes Game 1 from Game 2 with non-negligible advantage. We build a distinguisher $\mathcal{B}$ for decision-LWE.

The challenger for $\mathcal{B}$ gives a matrix

$$B \xleftarrow{\$} \mathbb{Z}_q^{n \times (m+1)}$$

and a vector

$$y \in \mathbb{Z}_q^{m+1},$$

where either

$$y = B^T r + e$$

for

$$r \xleftarrow{\$} \mathbb{Z}_q^n, \quad e \leftarrow \chi^{m+1},$$

or else y is uniform in $\mathbb{Z}_q^{m+1}$.

Parse

$$B = [A \mid h],$$

where $A \in \mathbb{Z}_q^{n \times m}$ and $h \in \mathbb{Z}_q^n$, and parse

$$y = (y_1, \dots, y_m, y_{m+1}) = (c_1, c'_2).$$

Then $\mathcal{B}$ gives the public key (A, h) to $\mathcal{A}$.

When $\mathcal{A}$ submits challenge messages μ_0, μ_1 , $\mathcal{B}$ picks $\beta \xleftarrow{\$} \{0, 1\}$ and returns the ciphertext

$$(c_1, c'_2 + \mu_\beta \Delta).$$

Now consider the two cases:

- If $y = B^T r + e$, then

$$c_1 = A^T r + e_1, \quad c'_2 = h^T r + e_2,$$

so the challenge ciphertext is distributed exactly as in Game 1.

- If y is uniform in $\mathbb{Z}_q^{m+1}$, then (c_1, c'_2) is uniform, and therefore

$$(c_1, c'_2 + \mu_\beta \Delta)$$

is distributed exactly as in Game 2.

Hence any distinguisher between Game 1 and Game 2 yields a distinguisher for decision-LWE. Therefore,

$$|\Pr[\text{Game}_1 = 1] - \Pr[\text{Game}_2 = 1]| \leq \text{Adv}^{\text{LWE}}(\lambda).$$

Final step. In Game 2, the challenge ciphertext is uniform, independent of β . Indeed, adding the fixed shift $\mu_\beta \Delta$ to a uniformly random element of $\mathbb{Z}_q$ preserves uniformity. Therefore the adversary gets no information about β , and

$$\Pr[\mathcal{A} \text{ guesses } \beta \text{ in Game 2}] = \frac{1}{2}.$$

Combining the three games, we get

$$\text{Adv}_{\mathcal{A}}^{\text{IND-CPA}} \leq \underbrace{|\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_1 = 1]|}_{\text{negligible by LHL}} + \underbrace{|\Pr[\text{Game}_1 = 1] - \Pr[\text{Game}_2 = 1]|}_{\text{negligible by LWE}},$$

which is negligible.

Hence, assuming decision-LWE is hard, the dual Regev public-key encryption scheme is

IND-CPA secure.